Hand Tremor Stabilizer

Project Report

Abstract

The **Smart Hand Tremor Stabilizer** is an embedded biomedical assistance device designed to reduce unwanted hand vibrations using a counter-vibration mechanism. The system detects hand tremors through the **GY-61 accelerometer sensor** and generates controlled out-of-phase vibrations using vibration motors to stabilize hand movement.

The project uses an ESP32 microcontroller, vibration motors, LEDs, push buttons, and an OLED display to create a compact and interactive stabilization system. The device is intended for research, assistive technology, and biomedical engineering applications.

Introduction

Hand tremors are involuntary rhythmic movements that can affect daily activities such as writing, holding objects, or operating tools. These tremors may occur due to neurological disorders, muscle fatigue, or aging.

The purpose of this project is to develop a wearable or portable stabilization system capable of minimizing tremors using vibration cancellation techniques. The system detects motion disturbances using the GY-61 sensor and produces controlled counter vibrations through motors to reduce unwanted movement.

This project demonstrates the application of embedded systems and motion control in biomedical assistive technologies.

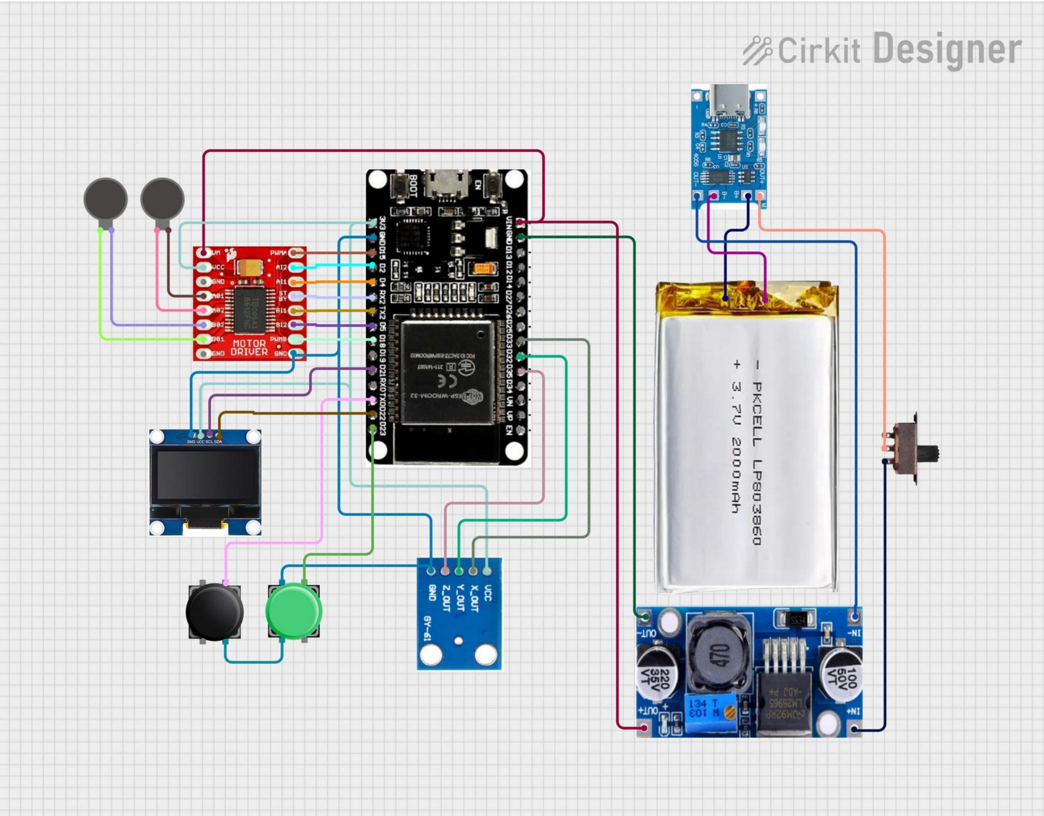
Objectives

The main objectives of this project are:

- To detect hand vibrations using the GY-61 accelerometer sensor.
- To generate counter vibrations using vibration motors.
- To reduce unwanted hand tremors through out-of-phase vibration control.
- To provide visual system status using LEDs and an OLED display.
- To create a simple user-controlled stabilization system.

Components Used

Component	Function
ESP32 Microcontroller	Main controller
GY-61 Accelerometer	Hand vibration detection
OLED Display SSD1306	System status display
Vibration Motors	Counter vibration generation
Motor Driver Module	Motor control
Push Buttons	User input
LEDs	Status indication



Working Principle

Tremor Detection

The GY-61 accelerometer continuously monitors hand movement and vibration patterns. When tremors are detected, the sensor sends motion data to the ESP32 microcontroller.

Counter Vibration Mechanism

The ESP32 processes the detected vibration signals and activates vibration motors through the motor driver circuit. The motors generate vibrations in the opposite phase of the detected tremor.

This out-of-phase vibration helps reduce the effect of unwanted hand movements and improves stabilization.

OLED Display Interface

The OLED display shows:

- Motor status
- LED status
- System operating mode
- Device activity information

The display provides real-time monitoring and user feedback.

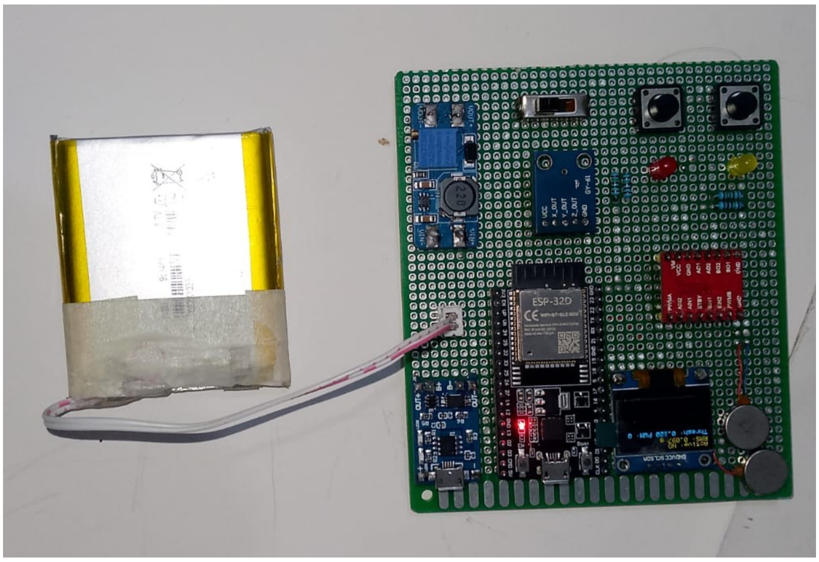
User Controls

Two push buttons are used for system operation:

Button	**Function**
Button 1	Toggle vibration motors
Button 2	Toggle LED indication

The LEDs indicate:

- Power and activity status
- Motor operation status



Software Description

The project is programmed using **Arduino C/C++** and includes the following libraries:

Library	**Purpose**
Wire.h	I2C communication
Adafruit_GFX.h	Graphics handling
Adafruit_SSD1306.h	OLED display control

PWM signals are used to control the speed and intensity of the vibration motors.

Features

- Hand tremor detection
- Counter vibration stabilization
- OLED display interface
- Real-time motor control
- User-friendly operation
- Compact embedded design
- Low-cost implementation

Advantages

- Helps reduce unwanted hand vibration
- Portable and lightweight
- Easy to operate
- Low power consumption
- Useful for biomedical assistive research
- Real-time stabilization response

Limitations

- Prototype-level stabilization accuracy
- Performance depends on sensor placement
- Limited medical validation
- External movement may affect readings

Applications

- Assistive healthcare devices
- Biomedical engineering projects
- Hand tremor stabilization research
- Wearable technology development
- Rehabilitation systems

Future Improvements

Future enhancements may include:

- Advanced motion analysis algorithms
- Adaptive vibration control
- Wireless connectivity
- Rechargeable battery system
- AI-based tremor prediction
- Wearable glove implementation

Conclusion

The **Smart Hand Tremor Stabilizer using Gyroscope-Based Counter Vibration System** demonstrates how embedded systems and motion sensors can be used to reduce unwanted hand tremors through controlled counter vibrations. The project successfully integrates vibration detection, motor control, and real-time monitoring into a compact and user-friendly system.

This project highlights the potential of smart assistive technologies for biomedical and rehabilitation applications.